

Network Design Brief
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In the case that a company is out of compliance, lacks the acquisition of a secure intranet, my goal and purpose are to design, establish, and secure the network infrastructure that meets the requirements of both the growing company and compliance standards.

Beginning with ensuring compliance with standard policies in the most production-efficient approach to engineering the intranet. First and foremost, a contract with a telecommunications service provider and an Internet service provider is required to establish an intranet to serve online customers. Generally, the most secure, cost-efficient choice is to bundle TSP and ISP services together from one vendor to streamline the intranet engineering phase.

- ❖ (NRA) Non-Routable Address initiation for the company's intranet will be their best foundational choice.
 - Inbound risks will be minimized, external traffic will be secured, and firewall resistance enhanced due to the non-routable private IP address, simplifying configuration procedures
 - Enforced regulations will be met by the installation of this service that sustains the availability, integrity, and security of both employees and customers.
 - The cost efficiency and processes are appropriate for the size of this organization.

Above are ISP standards and functioning components of NRA. I would like to clarify further NRA processes that are necessary for efficient production, design purposes, and organizational clarity by recommending the use of and explaining Active Address Translation

- ❖ (NAT) Network address translation
 - Incorporation of this is necessary to functionalize the mapping of the public IP NRA space.
 - Implementation will enable outbound service and further harden the security of the attack surface, amplify in-depth defense of inbound traffic, and seamlessly enhance organizational scalability.

Following and expanding upon the installation of an intranet. Regarding the internal intranet website and external website marketplace, specifically, the activation/configuration of required Domain Name Servers and Dynamic Host Name Configuration servers is intended for security, longevity, and compliance.

- ❖ Internal (DNS) Domain Name Server + (DHCP) Dynamic Host Name Configuration Protocol
 - Functioning separately, the DNS servers create clarity and security. with Active Directory file sharing, incident logging, data storage, and internal employee communication.

- Translation resolution of both unencrypted/encrypted data transmission handling and filtering enhances the scalability of both company production and security operations drastically.
- Defense against IP spoofing and ensures outgoing traffic reaches certified network targets.
- ❖ The Internal DHCP will automatically assign internal IPs, simplify device management
 - Enhancing and securing network control by assisting with the automated configurations of non-static IP processes.
 - Leased DHCP servers link with hostnames in DNS Servers, and manage DNS Server location.
 - Automated IP range configurations streamlining gateway determination and subnetting.

Once servers are operational, the Implementation of an Intrusion Prevention System and Intrusion Detection System is ideal at network choke points. Aligned to receive all wired and wireless data transmissions of both sensitive and non-sensitive consumer and employee data. While neither are regulatorily required to fulfill compliance, in tandem, they provide the necessary means to enforce that outcomes of production indeed do meet standardized policy-enforced procedures for PCI-DSS/NIST and other future regulations that may apply, considering company growth.

- ❖ (IPS) The Intrusion Prevention System is an active 24/7 component monitoring for the purpose of preventing bad acts and behavioral anomalies internally and externally.
 - Focused on customer transactions, customer and employee account changes, Ingoing and outgoing traffic for both the internal and marketplace websites.
 - Monitors employee data handling, internal communication, and outgoing communication, like emails or instant messaging
 - Defends against exploits and threat actor attempts.
- ❖ (IDS) The Intrusion Detection System is a passive 24/7 component monitoring for the purpose of detecting and logging typical internal and external network activity.
 - Categorical notification of flags: tracks DNS tunneling or suspicious query spikes.
 - Tracks and logs threat actor attempts like brute force attacks and successful intrusions, and post-intrusion lateral movement.
 - Detects abnormal behavior on “trusted” HTTP/HTTPS and DNS ports.
 - Flags can also include: Authentication failures, policy violations, Denial of Service, Data breaches, etc.

With proper protocols and procedures in place, usage of Windows Active Directory + Routing and Remote Access Service fulfills several company needs for employee identification, authentication, verification, and data handling. Providing ease of access to data transferring and account usage. Prerequisites for using AD are functioning internal DNS servers.

- ❖ Windows (AD) Active Directory contributes to the network identity plane by verifying employee credentials upon logging into host accounts, both locally and remotely, enabling remote work/production
 - Uses (SSO) Single sign-on for efficient access (MFA - Multi Factor Access suggested)
 - Allows for IT Security SRV logging and also the enforcement and delegation of the principle of least privilege. Increasing efficiency and clarity in handling employee data access permissions
 - Alternative logging source and/or efficient method to organize pre-existing logs and more.
- ❖ Windows (RRAS) Routing and Remote Access Service functions as a stand-alone Layer 3 router functionality, enabling VPN usage.
 - Provides legacy support for dial-up remote access if needed.
 - Employee remote access facilitates management of accounts and serves to sustain production.
 - Incorporates RADIUS proxy enforcing AAA protocols, strengthening IT Security.

Security & Privacy Policy Recommendations

- ☐ Enforce least privilege access
- ☐ Log DNS, DHCP, AD, and VPN activity for auditability
- ☐ Back up critical configs regularly
- ☐ Comply with relevant standards (e.g., PCI-DSS, NIST, GDPR, where applicable)
- ☐ Establish and consistently monitor firewall rules, IPS logs, and AD group memberships for anomalies.

Thank you for reviewing this preliminary Network-Infrastructure Design Brief.

I welcome your questions, feedback, or requests for further refinement—whether technical deep dives, budget scenarios, or phased roll-out schedules.

- ❖ Next steps:
 - ★ Stakeholder Alignment Meeting
 - Gather IT, Finance, and Compliance leads to validate that the proposed architecture, budget, and controls satisfy business targets and regulatory requirements.
 - Gap & risk workshop (1 hr) – Identify any blind spots and rank remediation urgency.
 - Road-map sign-off – Confirm scope, timeline, and success metrics.

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